

[Foundational Clinic 4] Grade 3-5 Order of Operations, Geometry Measurement & Word Problems

Student Name: _____

Target: _____

Time: 35 Mins Total: 100 Pts Score: _____

 Sophia &  William · BODMAS / Square Side from Area
/ Unit Scale / Speed Time

💡 Key Rules & Common Mistake Prevention:

① **Order of Operations (BODMAS):** Brackets → Orders/Powers → Division & Multiplication (left to right) → Addition & Subtraction (left to right). Never add brackets randomly or do addition before multiplication without brackets!

② **Square Area vs. Side Length:** $\text{Area} = \text{Side} \times \text{Side} = s^2$. To find side length from area: $s = \sqrt{\text{Area}}$, **NEVER** $\text{Area} \div 2$! If $\text{Area} = 36 \text{ cm}^2$, $\text{side} = 6 \text{ cm}$ (since $6 \times 6 = 36$), NOT 18 cm!

③ **Metric Volume & Area Scale Factors:** $1 \text{ m} = 100 \text{ cm} \Rightarrow 1 \text{ m}^2 = 10,000 \text{ cm}^2 \Rightarrow 1 \text{ m}^3 = 1,000,000 \text{ cm}^3$. $1 \text{ Liter} = 1,000 \text{ mL} = 1,000 \text{ cm}^3$.

④ **Speed, Distance, Time Units:** 40 minutes is $\frac{40}{60} = \frac{2}{3}$ of an hour, **NEVER 0.4 hours!** When speed is in km/h, time must be in hours!

Section 1: Order of Operations & Mental Arithmetic Rigor (4 pts each, 24 pts total) Score: _____

Calculate step-by-step strictly following BODMAS. Do NOT skip intermediate steps:

(a) $18 + 5 \times 4 =$ _____

Multiply first! (Check: $18 + 5 \neq 90$!) _____

(b) $(18 + 5) \times 4 =$ _____

Brackets first! _____

(c) $48 \div 6 \times 2 + 3^2 =$ _____

Powers first, then division/mult left-to-right. _____

(d) $50 - 2 \times (12 - 3 \times 2) =$ _____

Inside the brackets: 3×2 first. _____

(e) $24 \div (2 + 6) \times 5 =$ _____

Brackets first: $(2 + 6) = 8$. _____

(f) $100 - [40 - (15 + 5)] \div 4 =$ _____

Innermost brackets first. _____

Section 2: Perimeter, Area & Square Side Recovery (4 pts each, 20 pts total) Score: _____

Pay close attention to units and formulas. Recall: $\text{Area of square} = s^2$, $\text{Perimeter} = 4s$:

(a) A square has an area of 49 cm^2 . _____

Side length = _____ cm (NOT $49 \div 2$!) _____

(b) A square has an area of 81 cm^2 . _____

What is its perimeter? _____

(c) A rectangle has perimeter 36 cm and width 6 cm. _____

Length = _____ cm; Area = _____ cm^2 . _____

(d) A square garden has perimeter 48 m. _____

What is its area in m^2 ? _____

(e) A wire of length 40 cm is bent into a rectangle with width 8 cm. Another identical wire is bent into a square. _____

Which shape has the greater area, and by how many cm^2 ? _____

Section 3: Metric Measurement & Scale Factor Conversions (3 pts each, 18 pts total) Score: _____

Convert accurately. Remember: 1D is $\times 10^1$, 2D is $\times 10^2$, 3D is $\times 10^3$:

(a) $3.5 \text{ m} = \underline{\hspace{2cm}} \underline{\hspace{2cm}}$
cm

(b) $2.4 \text{ m}^2 = \underline{\hspace{2cm}} \underline{\hspace{2cm}}$
 cm^2

(c) $0.05 \text{ m}^3 = \underline{\hspace{2cm}} \underline{\hspace{2cm}}$
 cm^3

(d) $4,500 \text{ mL} = \underline{\hspace{2cm}} \underline{\hspace{2cm}}$
L

(e) $1.8 \text{ L} = \underline{\hspace{2cm}} \underline{\hspace{2cm}}$
 cm^3

(f) $150 \text{ minutes} = \underline{\hspace{2cm}} \underline{\hspace{2cm}}$
hours

Section 4: Speed, Distance & Time Precision (6 pts each, 18 pts total) Score: _____

1. [Time Conversion & Distance]

A school bus travels from SUIS Hongqiao to the sports stadium at an average speed of 45 km/h . The trip takes 40 minutes.

- (a) Express 40 minutes as a fraction of an hour in simplest form: _____ hour.
(b) How many kilometers did the bus travel? Show formula and working:

Distance = Speed $\times$ Time =

2. [Speed Comparison]

Sophia runs 600 meters in 2 minutes 30 seconds. William cycles 3.6 kilometers in 12 minutes.

- (a) Find Sophia's speed in meters per second (m/s).
(b) Find William's speed in meters per second (m/s). Who is faster?

Working:

Section 5: Structured Problem-Solving Clinic (10 pts each, 20 pts total) Score: _____

1. [Water Tank Capacity & Volume]

A rectangular aquarium has a base measuring 80 cm long and 50 cm wide. It contains water to a depth of 30 cm.

- (a) Calculate the volume of water in the tank in cm^3 .
(b) Convert this volume to Liters ($1 \text{ L} = 1,000 \text{ cm}^3$).
(c) If a decorative rock with volume $8,000 \text{ cm}^3$ is submerged completely in the aquarium, by how many centimeters will the water level rise?

Structured 3-Step Working:

2. [Money & Ratio Allocation]

Sophia and William received a total of ¥360 for helping set up the SUIS Science Fair. Sophia worked for 5 hours and William worked for 4 hours. They agreed to split the money in proportion to the hours worked.

- (a) How much money does each person receive per hour?
(b) How much money does Sophia receive, and how much does William receive?
(c) Check: Does the sum of their shares equal ¥360?

Working & Reality Check:



Complete Answer Key & Pedagogical Notes

Section 1 (BODMAS):

- (a) $18 + 20 = 38$ (not 90!); (b) $23 \times 4 = 92$;
(c) $8 \times 2 + 9 = 16 + 9 = 25$;
(d) $50 - 2 \times (12 - 6) = 50 - 2 \times 6 = 50 - 12 = 38$;
(e) $24 \div 8 \times 5 = 3 \times 5 = 15$;
(f) $100 - [40 - 20] \div 4 = 100 - 20 \div 4 = 100 - 5 = 95$.

Section 3 (Units):

- (a) 350 cm; (b) $2.4 \times 10,000 = 24,000 \text{ cm}^2$;
(c) $0.05 \times 1,000,000 = 50,000 \text{ cm}^3$;
(d) 4.5 L; (e) $1,800 \text{ cm}^3$; (f) $150 \div 60 = 2.5$ or $2\frac{1}{2}$ hours.

Section 2 (Geometry):

- (a) Side = $\sqrt{49} = 7 \text{ cm}$ ($7 \times 7 = 49$, not 24.5!);
(b) Side = $\sqrt{81} = 9 \text{ cm} \implies \text{Perimeter} = 4 \times 9 = 36 \text{ cm}$;
(c) Length + 6 = $36 \div 2 = 18 \implies \text{Length} = 12 \text{ cm}$; Area = $12 \times 6 = 72 \text{ cm}^2$;
(d) Side = $48 \div 4 = 12 \text{ m} \implies \text{Area} = 12^2 = 144 \text{ m}^2$;
(e) Rect: $l = 20 - 8 = 12 \text{ cm}$, Area = $12 \times 8 = 96 \text{ cm}^2$. Square: side = 10 cm, Area = 100 cm^2 . The square has greater area by 4 cm^2 .

Section 4 & 5 (Speed & Applications):

- Sec 4: 1. (a) $\frac{40}{60} = \frac{2}{3} \text{ hr}$; (b) $45 \times \frac{2}{3} = 30 \text{ km}$.
2. (a) $2 \text{ min } 30 \text{ s} = 150 \text{ s} \implies 600 \div 150 = 4 \text{ m/s}$.
(b) $3,600 \text{ m} \div (12 \times 60 \text{ s}) = 3,600 \div 720 = 5 \text{ m/s}$. William is faster.
Sec 5: 1. (a) $80 \times 50 \times 30 = 120,000 \text{ cm}^3$; (b) $120,000 \div 1000 = 120 \text{ L}$; (c) Base Area = $80 \times 50 = 4,000 \text{ cm}^2$. Rise = $8,000 \div 4,000 = 2 \text{ cm}$.
2. (a) Total hours = $5 + 4 = 9 \text{ h}$. Rate = $360 \div 9 = \text{¥}40/\text{h}$; (b) Sophia: $5 \times 40 = \text{¥}200$; William: $4 \times 40 = \text{¥}160$; (c) $200 + 160 = 360$ (Verified).